

Feedback on the Revised Draft

1. Overview

The revision is substantial and the work has clearly improved. Four of the five required revisions from my earlier feedback are fully closed, and both optional enhancements were adopted — the relocation of the elementary transformer mechanics to Appendix C in particular has rebalanced the report considerably in favour of results and analysis, which is where an examiner will spend their attention.

Two of the closures went beyond what was asked. The asymmetric loss correction in §3.3 now explains why the weight indexes on the ground-truth class rather than simply restating the formula, and the reordered Future Work list in §6.1 carries an explicit impact rationale for each item. Both read as genuine engagement rather than compliance.

What follows is a verification of each item from my earlier feedback, then a short set of remaining points to address before the Milestone 6 submission. None of them require new analysis of the primary model, and none change the report's conclusions.

2. Verification Against My Earlier Feedback

Item from my earlier feedback	Status	Evidence in the revised report
(a) Table of Contents before Abstract, with chapters, sub-topics and page numbers	CLOSED	Field-generated TOC now appears on page 2, ahead of the Abstract. Depth extends to third-level entries (3.4.0–3.4.3, C.1–C.2) and matches the body headings exactly. See the related note in Section 3 below regarding page numbering in the body.
3.1 Asymmetric loss formula notation	CLOSED	Section 3.3 now states the standard weighted cross-entropy over the one-hot true label, and adds an "Important implementation detail" paragraph explaining that <code>nn.CrossEntropyLoss(weight=...)</code> indexes the weight vector by ground-truth class. The one-hot collapse to $-w_x \log(p_x)$ is shown explicitly. This exceeds the minimum correction requested.
3.2 ProtectAI DeBERTa missing latency	CLOSED	Option B adopted. The footnote appears against Table 4.1, the Table 4.4 leaderboard, and the Table 5.1 comparison, so no table now presents an unexplained dash. See Section 3 for a recommended strengthening.
3.3 Appendix B OOD F1 notation inconsistency	CLOSED	All instances now use percentage format — 33.68% in the Abstract, §3.7, §4.6, §5.4, Conclusions and Appendix B. The contract target is stated consistently as $\geq 70.00\%$ throughout.
3.4 Section 3.5 KNN deviation — comparative sentence	OPEN	Section 3.5 Paradigm A still offers only a conceptual justification (geometric nearest-neighbour semantics; tree ensembles do not implement distance-based retrieval). Neither the quantitative comparison nor the literature

		reference I requested has been added. Detail in Section 3.1 below.
Optional — Section 2 length	ADOPTED	The elementary embedding-layer and multi-head self-attention material has been relocated to a new Appendix C, with a forward pointer retained in §2.2. Section 2 is now roughly 9% of body text, down from approximately one third. The running "Ignore previous instructions" example and the end-to-end pipeline summary (now §2.4) both survived the move intact.
Optional — Section 6.1 ordering	ADOPTED	Future Work is now explicitly ordered by expected impact on the two unmet contract targets, with a lead-in sentence stating the ordering logic and an impact tag on each of the five items.

3. Items to Address Before Final Submission

3.1 Section 3.5 — KNN deviation still requires a comparative sentence (carried over)

The situation. The contract said Paradigm A (the non-transformer comparison) would use Random Forest or XGBoost on top of the embeddings. You used K-Nearest Neighbours instead. That's a deviation from the contract, and a deviation is fine as long as it's justified.

What you wrote as justification. "KNN is what a vector-database paradigm means; Random Forest isn't a distance-based method." That doesn't work as a justification, because the contract is the thing that defined what Paradigm A was — and it said Random Forest. You are essentially saying "I didn't do what the contract said because what the contract said isn't what this is." That's the circularity.

So I am asking you one thing: give a real reason for using KNN instead of Random Forest. Either

- run Random Forest on the embeddings you already have and report the numbers — then the reason becomes "I tried both, here's why KNN was the better comparison," or
- if there's no time, cite a source supporting the claim that tree methods work poorly on sentence embeddings

There is a **second and sharper issue** underneath it. Conclusion 2 — that asymmetric loss is the differentiator, because vector search and generative judges cannot encode an explicit false-positive penalty — is the strongest research claim in the report, and as stated it is vulnerable. Scikit-learn's RandomForestClassifier accepts a class_weight parameter, and KNN predictions can be thresholded on neighbour-vote proportions rather than decided by majority vote. Both can be made false-positive averse. **The open question an examiner can therefore ask is whether Paradigm A's 21.92% FPR is a property of non-transformer embedding classifiers in general, or only of the specific unweighted KNN configuration that was run.**

Running Random Forest over the MiniLM embeddings resolves both problems in a single pass, and the cost is genuinely low: `vector_db_classifier.py` already computes `train_embeddings`, so what remains is a fit, a predict, and a timing loop over the same frozen test set. Either outcome is reportable:

- **If Random Forest is comparable on F1 but slower or worse on FPR:** the contract deviation becomes justified on evidence rather than definition, and Conclusion 2 is strengthened.
- **If Random Forest with `class_weight={0:3, 1:1}` clears the FPR constraint:** that is a more interesting finding than the one currently in the report. Conclusion 2 would then need narrowing to something like “off-the-shelf vector search as deployed here does not encode a false-positive penalty” — still true, and considerably more defensible under questioning.

If time does not permit the run, the literature fallback remains acceptable, but it should make a real argument rather than a gesture. A serviceable form: axis-aligned splits over 384 arbitrary latent dimensions are a poor inductive fit for embeddings whose semantic signal resides in cosine geometry, citing Reimers & Gurevych [15], which is already in the reference list. One sentence either way closes the contract obligation.

3.2 Section 4.1 — ProtectAI latency: the footnote is acceptable, but a measurement is strongly preferable

The footnote satisfies the letter of what I asked for, and no table now carries an unexplained dash, so this item is formally closed. I am reopening it as a recommendation rather than a requirement, for the following reason.

The contract criterion is a latency advantage of at least 25% over the baseline. The footnote discharges that with an argument from parameter count — 138M against 66M — which is an architectural expectation, not a measurement, and Appendix B currently records the comparative-baseline requirement as satisfied partly on that basis. An examiner is entitled to treat the claim as unverified, and the exchange that follows will be harder to handle than the fifteen minutes the measurement would cost.

I checked `results/master_benchmark_report.json` in the repository: the ProtectAI entry carries accuracy, precision, recall, F1, FPR and the full confusion matrix, but has no `avg_latency_ms` field, while all three other models do. The figure was never captured, so it cannot be retrieved from the existing artifacts. This has to be a short re-run or nothing.

If you do re-run it, the protocol must match the one used for DistilBERT or the comparison is not valid. From `evaluate_custom_model.py`, that means: MPS device, the text-classification pipeline with `truncation=True` and `max_length=512`, `batch_size=1` (set explicitly to simulate sequential user requests), and wall-clock across all 1,104 frozen test prompts divided by 1,104, on the same machine. Two refinements I would add beyond the existing script:

- Discard a few warm-up passes before starting the clock. First-call graph compilation on MPS inflates the mean noticeably on a run of this length.
- Report median alongside mean. The long-prompt tail in this corpus skews a wall-clock average, and the median is the more honest per-request figure.

Expect a result somewhere in the 25–40 ms range — most likely still inside the 50 ms contract ceiling, but three to four times DistilBERT’s 10.9 ms, which converts the latency-advantage claim from assumed to decisive. Substituting the real number into Tables 4.1, 4.4 and 5.1 would also let you delete the footnote entirely, which resolves §3.3 below at the same time.

3.3 Page numbering is absent from the document body

The Table of Contents displays page numbers, but the pages themselves carry none. On inspection the document contains no footer definition at all, and the only page-number fields anywhere in the file are the internal cross-reference fields inside the TOC itself. The effect is that the TOC is decorative rather than navigable — a reader told that the BIPIA results are on page 21 has no way to find page 21.

Insert → Page Number → Bottom of Page resolves it. Do this before generating the submission PDF, and refresh the TOC afterwards (right-click → Update Field → Update entire table) so the entries reflect the final pagination after the other revisions land.

3.4 Repository and reference housekeeping

- **Baseline evaluation script is not committed.** The repository `src/` directory contains the training script, the custom-model evaluation, both alternative paradigms, the benchmark aggregator, the sweeps and the BIPIA protocol — but no ProtectAI baseline evaluation script, and Appendix A lists the same set. Milestone 2b is therefore not reproducible from the repository, which slightly overstates the “reproducible artifacts” line in the scorecard, and is very likely why the latency figure was lost. Committing that script and adding it to the Appendix A table closes the gap.
- **Three identical footnotes.** The ProtectAI latency caveat is repeated verbatim as three separate footnotes against Tables 4.1, 4.4 and 5.1. Keep the full text once at Table 4.1 and have the other two cross-reference it — or delete all three if the measurement in §3.2 is taken.
- **Heading formatting drift.** Section 3.6 and the Appendix C sub-headings (C.1, C.2) are not italicised in the style used by their sibling headings throughout the rest of the report. Cosmetic, but visible in a printed copy.
- **Appendix C carries no citations.** The relocated material includes the scaled dot-product attention formula and the embedding-matrix walkthrough, neither of which is now attributed. Reference [1] (Vaswani et al.) belongs in §C.2 at minimum.